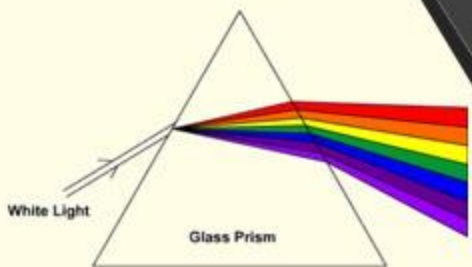


PHYSICS

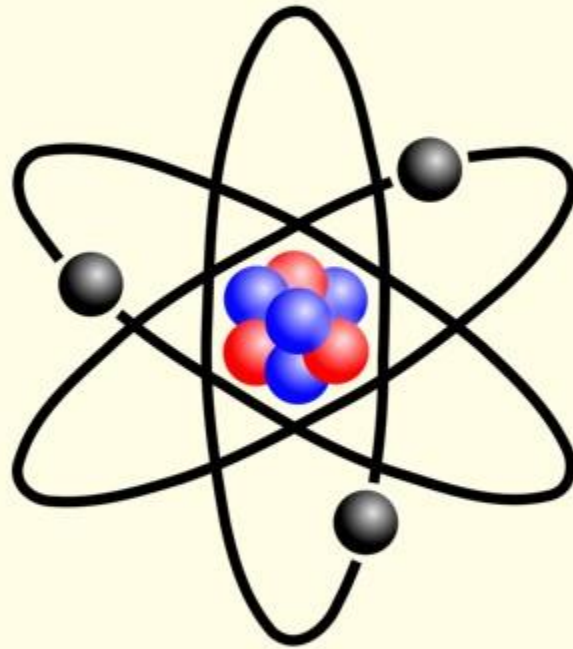
ELECTRICITY

MODULE - 15





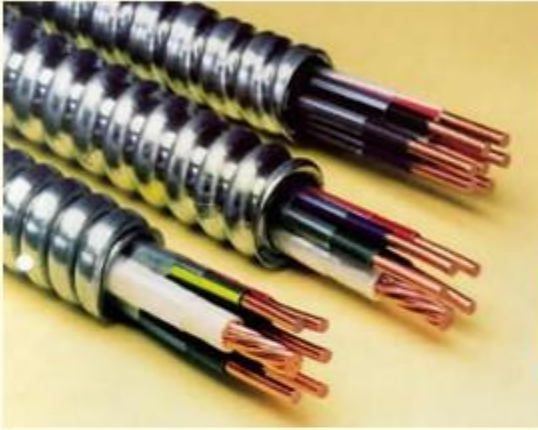
CHARGES AROUND US



- Every matter is made up some charges like protons and electrons.
- A Proton has a positive charge of $+1.6 \times 10^{-19}$ Coulomb
- A Electron has a negative charge of -1.6×10^{-19} Coulomb
- Like charges repel each other whereas unlike charges attract each other.



CONDUCTORS AND INSULATORS

**1**

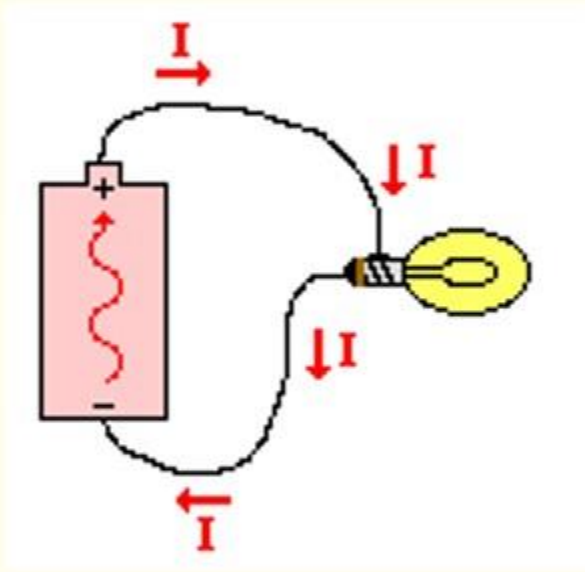
- **Conductors** are those materials in which electrons are free to move anywhere inside the matter.
- All metals are conductors because of the free electrons.

2

- **Insulators** are those materials in which electrons are not free to move anywhere.
- They can be plastic, wood or glass.



WHAT DO YOU MEAN BY ELECTRIC CURRENT?



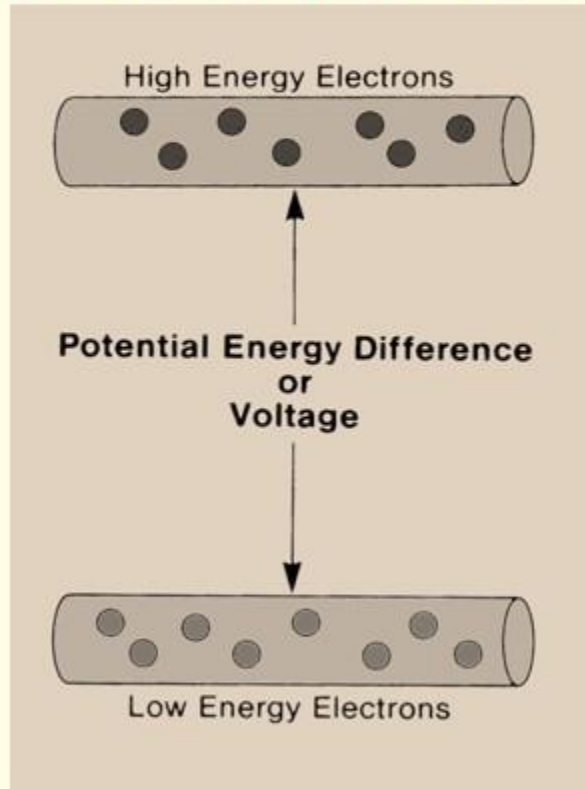
- The particles that carry charge through wires in a circuit are mobile electrons.
- The direction of an electric current is by convention the direction in which a positive charge would move. Thus, the current in the external circuit is directed away from the positive terminal and toward the negative terminal of the battery.
- Electrons would actually move through the wires in the opposite direction
- Thus, the number of charges flowing through a cross-section of a conductor per unit time is called current.

Electric current is directed from positive to negative terminal





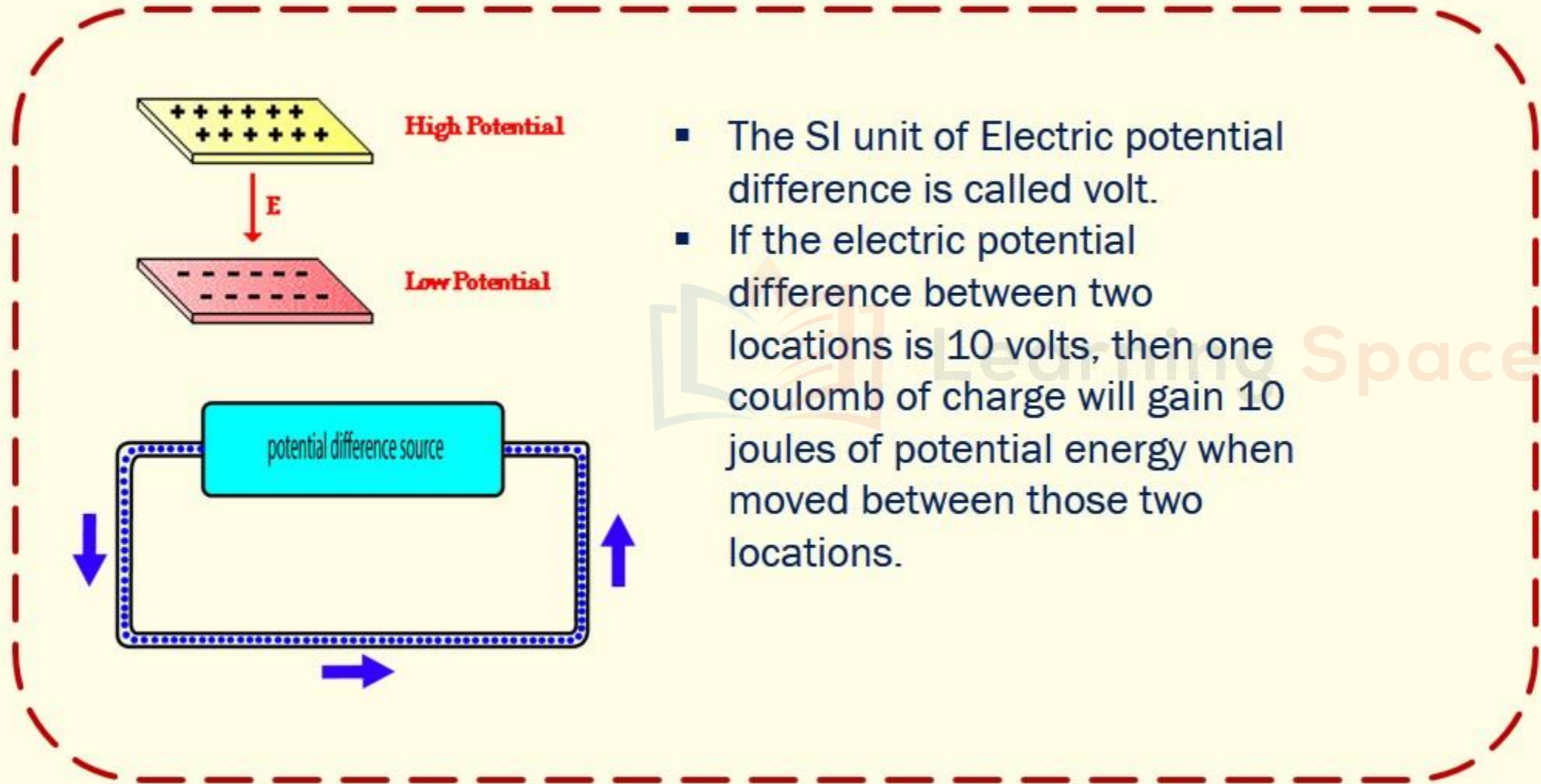
WHAT DO YOU MEAN BY ELECTRIC POTENTIAL?



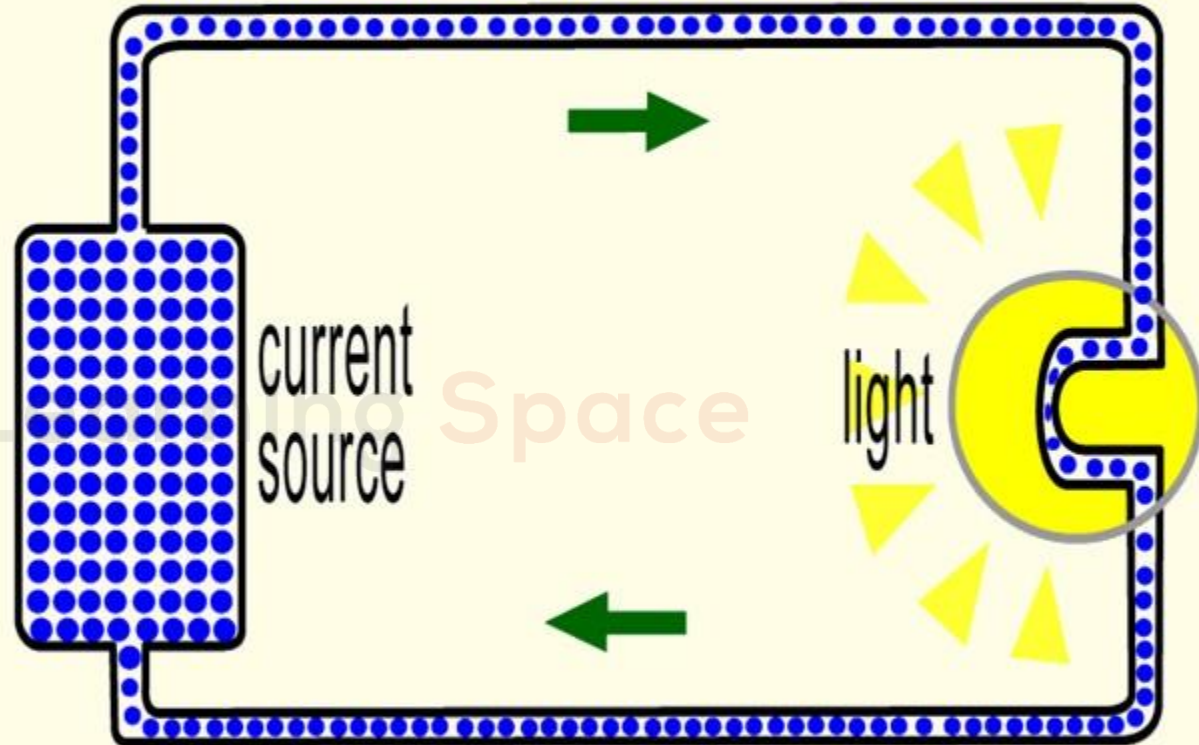
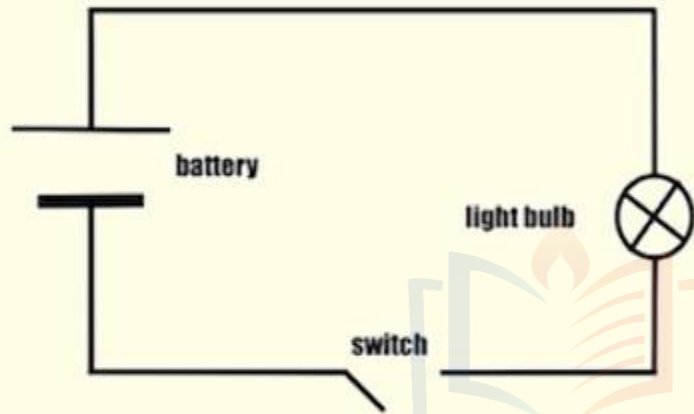
- When a Coulomb of charge possesses a relatively large quantity of potential energy at a given location, then that location is said to be a location of high electric potential.
- Similarly, if a Coulomb of charge possesses a relatively small quantity of potential energy at a given location, then that location is said to be a location of low electric potential.
- The difference between the potential or the potential difference which moves the charge from one potential to another.



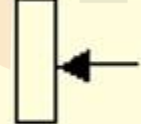
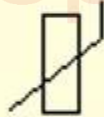
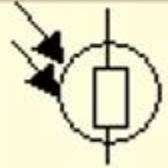
WHAT IS THE 'VOLT'?



A SIMPLE ELECTRIC CIRCUIT

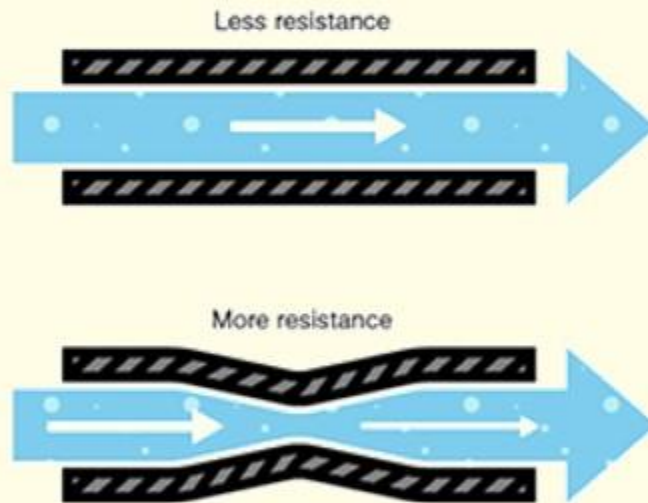


COMPONENTS OF AN ELECTRIC CIRCUIT

	Ammeter		Cell		Resistor
	Voltmeter		Battery		Variable resistor
	Diode		Potentiometer		Thermistor
	Light emitting diode		Light depended resistor		



WHAT DO YOU MEAN BY ELECTRICAL RESISTANCE?

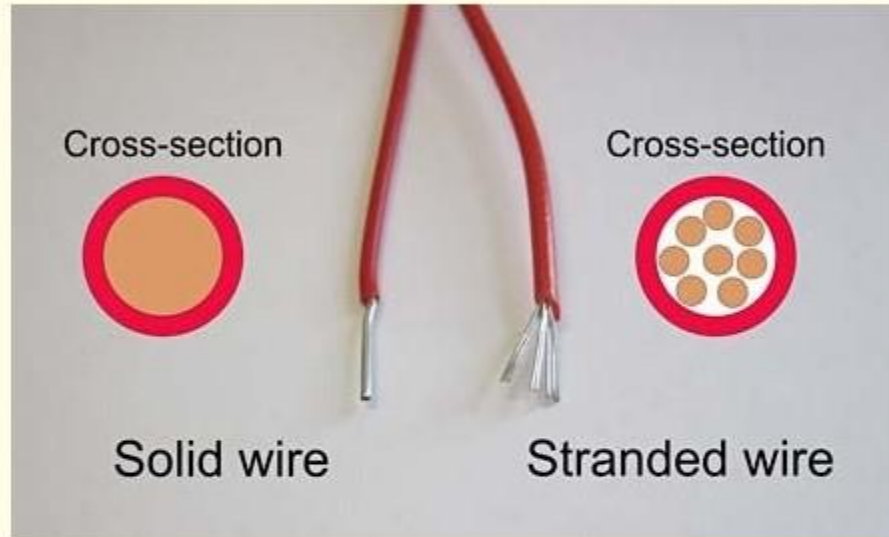


- An electron traveling through the wires encounters resistance.
- It is the hindrance to the flow of charge.
- While the electric potential difference established between the two terminals encourages the movement of charge, it is resistance that discourages it.
- It is represented by Ohm Ω





FACTORS ON WHICH RESISTANCE DEPENDS



- 1) Nature of the material or its resistivity
- 2) The length of the conductor
- 3) The thickness or area of cross-section of the conductor.

$$R = \frac{\rho L}{A}$$

ρ = resistivity

L = length

A = cross sectional area





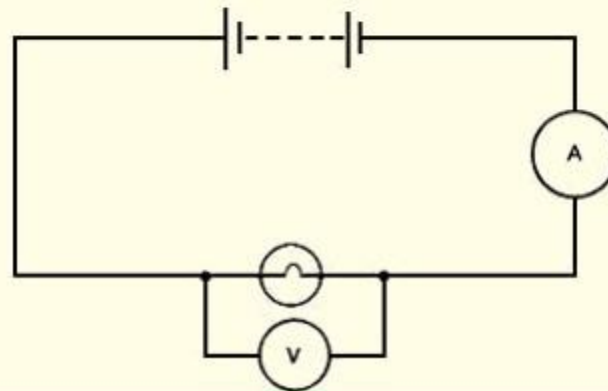
Measuring 'I' and 'V' in a Circuit

Ammeter

- It is always connected in series.
- Resistance of an ideal ammeter is zero
- This is because there is no voltage drop across it, it won't affect a circuit and would not change the current in the circuit.

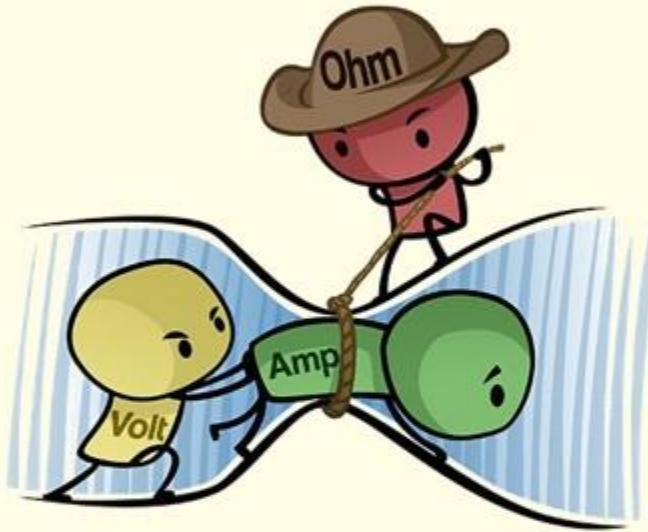
Voltmeter

- It is always connected in parallel
- Resistance of an ideal voltmeter is infinity
- This is because it draws no current from the circuit and can only indicate the voltage it is connected across

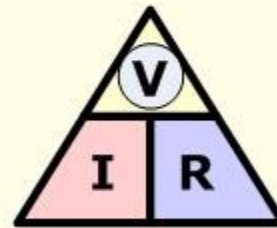




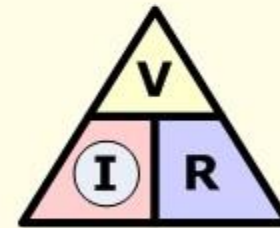
THE OHM'S LAW



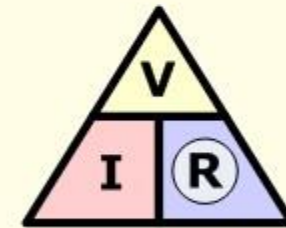
Ohm's law states that the current through a conductor between two points is directly proportional to the voltage across the two points. Introducing the constant of proportionality, the resistance



$$\textcircled{V} = I \times R$$



$$\textcircled{I} = \frac{V}{R}$$



$$\textcircled{R} = \frac{V}{I}$$





OHM'S LAW - APPLICATIONS

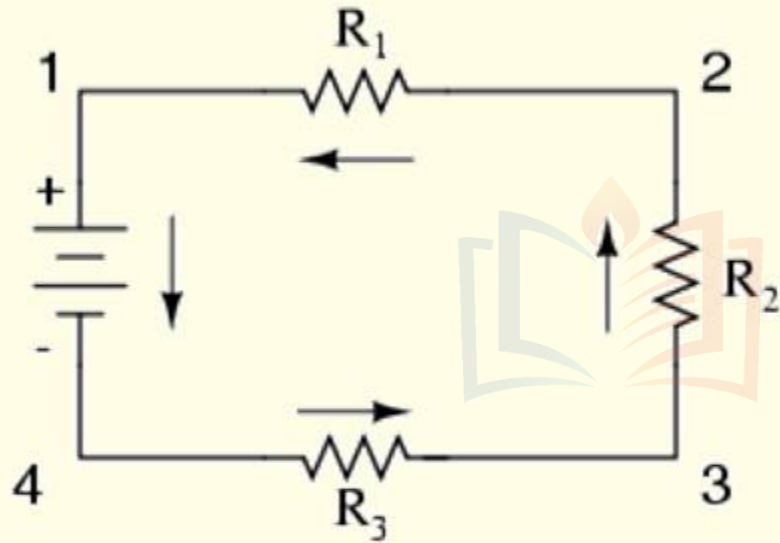


- 1) The power supplied to any electric heater can be calculated using this law. Power is Electric current times the Voltage.
- 2) From Ohm's law we can get the amount of heat produced by the coils of the device.
- 3) The speed of the conventional fans can be controlled using a potentiometer is also an application of this law.
- 4) Mobile phone & laptop chargers use DC power supply in operations. The working of this supply depends on the Ohm's law.

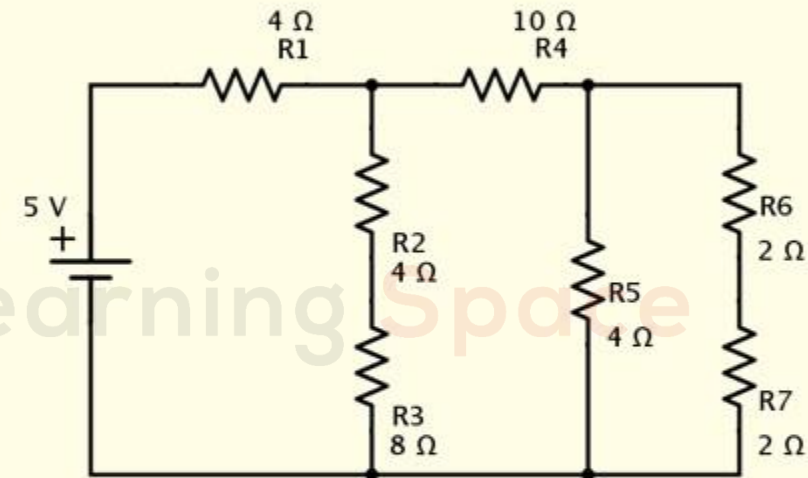




SERIES AND PARALLEL COMBINATION



$$R_{\text{total}} = R_1 + R_2 + R_3$$

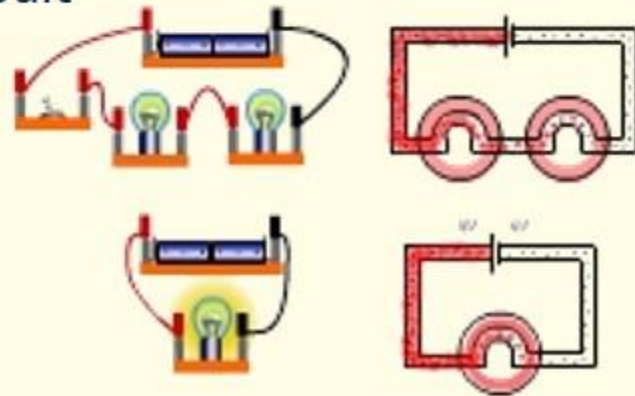


$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

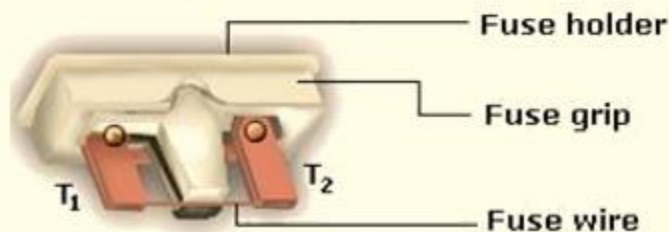


PARALLEL COMBINATION- ADVANTAGES

- 1) In series combination, either all devices are ON or OFF,
 - 2) In series connection, if one device fails, the entire circuit breaks,
 - 3) In series connection, all devices get same current. This is harmful for smaller devices.
 - 4) Voltage decreases for subsequent devices, so not useful for a domestic circuit
- 1) In parallel combination devices can be selectively switched ON/OFF
 - 2) In parallel connection others remain connected even if one circuit breaks.
 - 3) In parallel connection all devices get different current so its very useful for proper working
 - 4) Voltage is constant so useful for a domestic circuit.



HEATING EFFECT OF ELECTRIC CURRENT



- 1) The heating of the resistor is commonly known as Heating Effect of Electric Current.
- 2) This happens when a current continuously passes through the coils of the resistor and the chemical energy of the electric charges gets converted to electrical energy.
- 3) This heat is a waste product generated when current electricity is passed through a coil.
- 4) But it is also used in the working of heating appliances like
 - Electric heater
 - Toaster
 - Electric Iron
 - Hair straightener
 - Electric fuse

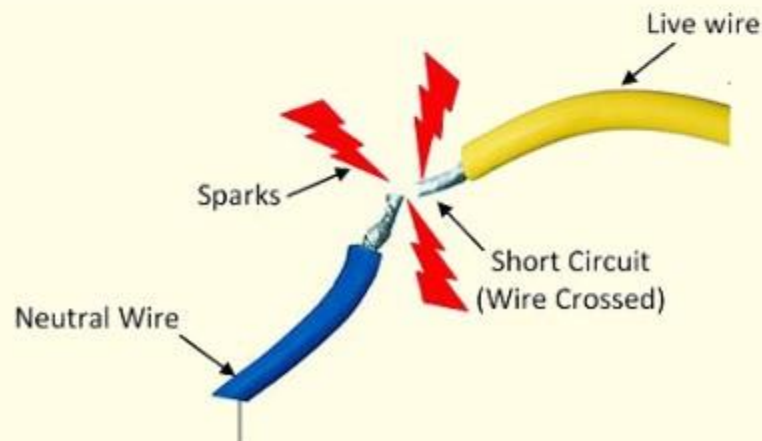
WHAT IS POWER IN A CIRCUIT?



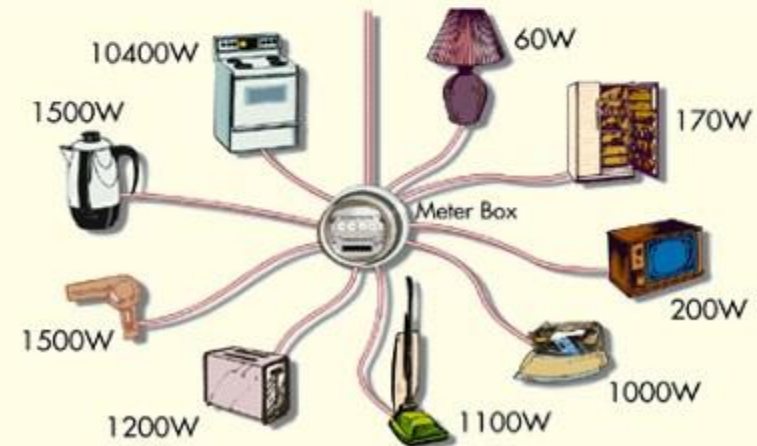
- The rate at which electric energy is consumed is called Electric power.
- SI unit of Electric Power is called Watt.
- Power is used for rating of the device such as a 100 W bulb etc.
- Higher Units are Kilowatt, Megawatt etc.
- Units of Energy for domestic use are denoted by Kilowatt Hour.
- When we pay the Electricity Board, we pay for the units of electricity consumed.
- The units are the measured in Energy Kilowatt Hour and not Power.

WHAT IS SHORT CIRCUITING AND OVER LOADING ?

- When two live wires gets unintentionally connected, the current takes a easy path of lower resistance.
- As resistance decreases, Current travelling in it increases, and it causes excess heat that is seen as sparks.
- This phenomenon can damage the inner circuit of the other connected appliances due to heavy current flow.



- The current in a circuit depends on the rating of the appliances connected to it. The choice of wires depends upon the maximum current estimated to pass through them
- If the total power rating of the appliances exceeds this permitted limit, they tend to draw a large current. This is known as overloading.



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